



Machine Learning and the Physical World

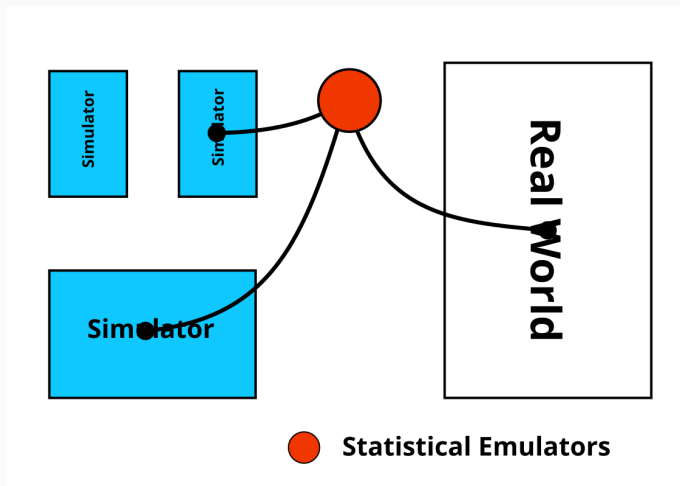
Lecture 9 : Reinforcement Learning

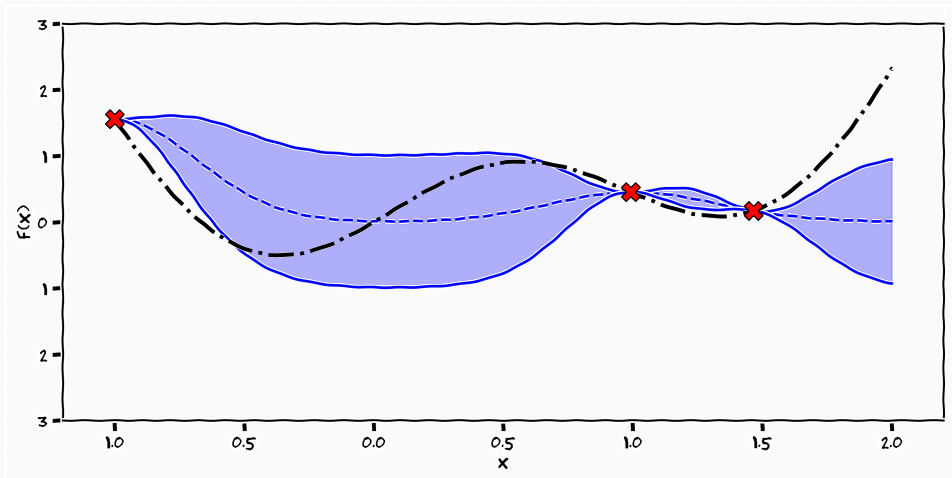
Carl Henrik Ek - che29@cam.ac.uk

10th of November, 2025

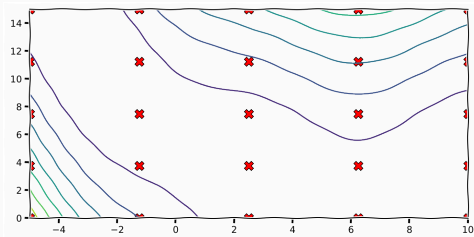
<http://carlhenrik.com>

Simulation and Emulation





Branin Function

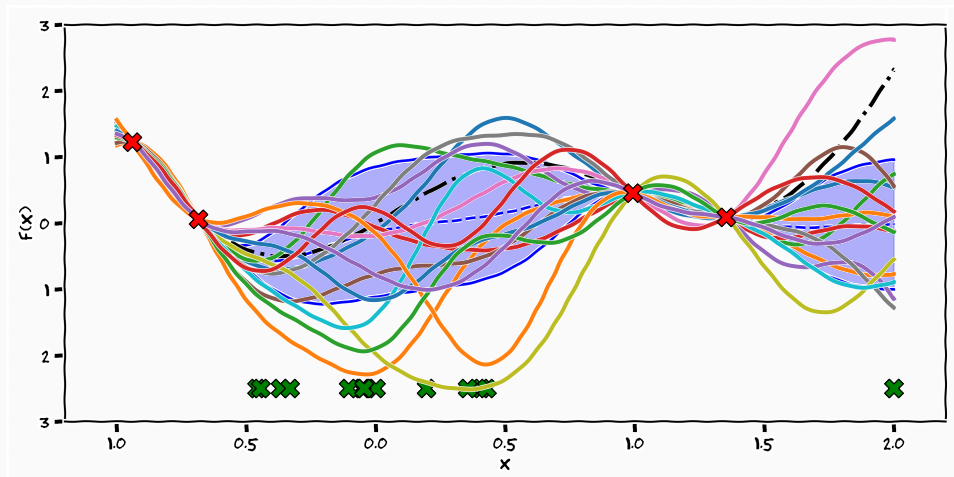


Fit a GP surrogate

$$k(\mathbf{x}_i, \mathbf{x}_j) = k_{\text{Exp}}(\mathbf{x}_i, \mathbf{x}_j) + k_{\text{White}}(\mathbf{x}_i, \mathbf{x}_j) + k_{\text{Bias}}$$

$$\mathbb{E}[f(U)] \approx \mathbb{E}[p(\mathbf{f} \mid \mathbf{Y}, \mathbf{X}, \mathbf{X}_{\text{samples}})] = 70.25$$

Surrogate Uncertainty

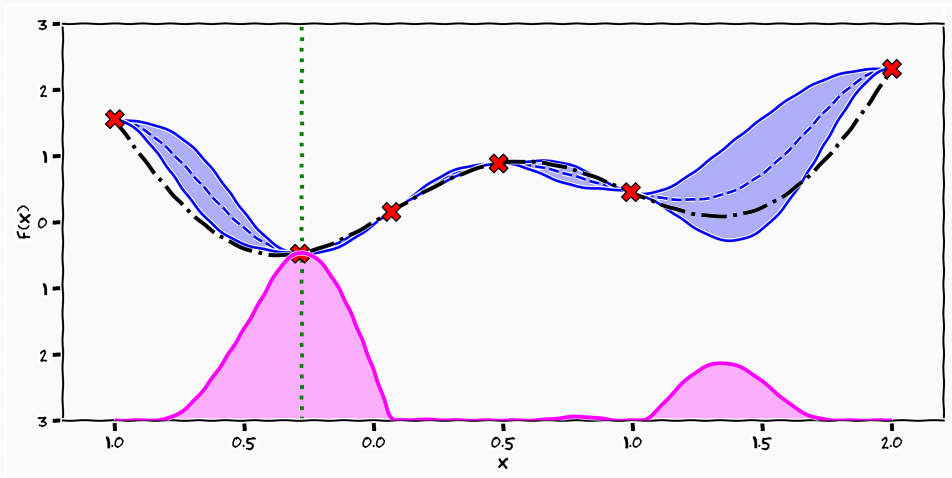


- Define a function that defines the **utility** of observing each location

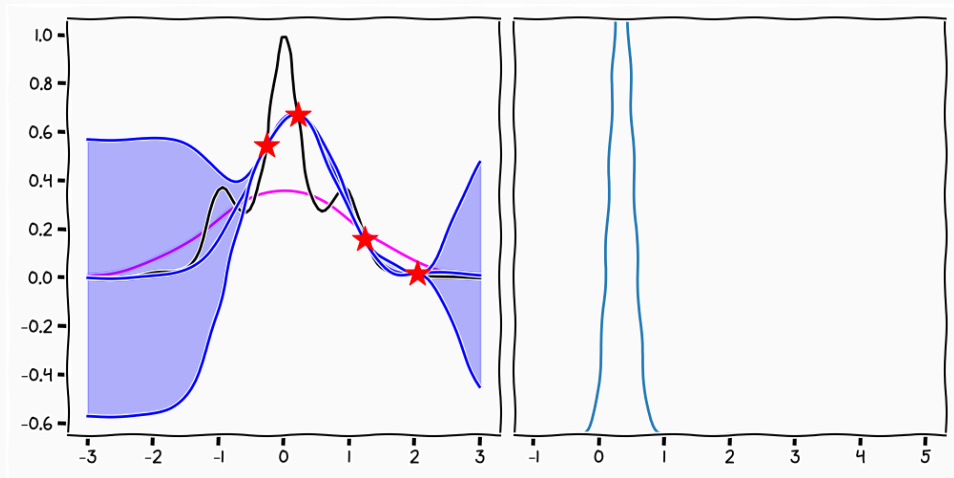
$$u(x, f(x^{(*)}), \mathcal{M}_n)$$

- Define the acquisition function as the expected utility

$$\begin{aligned}\alpha(x; \{x_i, y_i\}_{i=1}^n, \mathcal{M}_n) &= \mathbb{E}_{p(f)}[u(x)] \\ &= \int u(x, f(x^{(*)}), \mathcal{M}_n) p(f \mid \{x_i, y_i\}_{i=1}^n) df\end{aligned}$$



Quadrature



November 10, 2025

Reinforcement Learning

Supervised Learning predict output from input

$$\mathcal{D} = \{y_i, x_i\}_{i=1}^N$$

$$p(\mathbf{y}|\mathbf{x}) = \int p(\mathbf{y}|\mathbf{x}, \theta)p(\theta)d\theta$$

Supervised Learning predict output from input

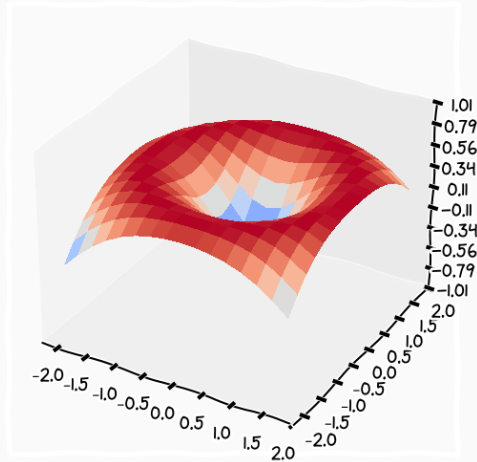
$$\mathcal{D} = \{y_i, x_i\}_{i=1}^N$$

$$p(\mathbf{y}|\mathbf{x}) = \int p(\mathbf{y}|\mathbf{x}, \theta)p(\theta)d\theta$$

Unsupervised Learning model the data

$$\mathcal{D} = \{y_i\}_{i=1}^N$$

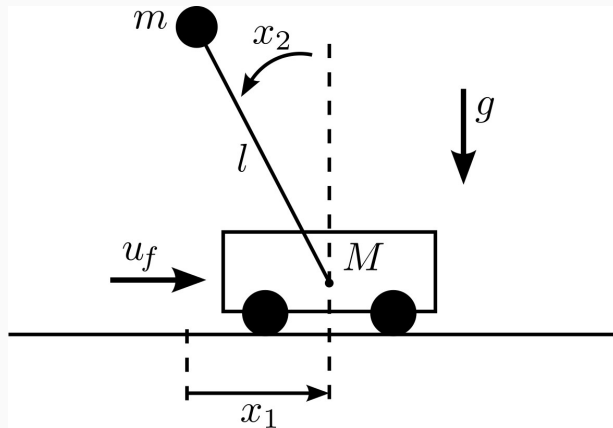
$$p(\mathbf{y}) = \int p(\mathbf{y}|\mathbf{x}, \theta)p(\mathbf{x})p(\theta)d\mathbf{x}d\theta$$



Reinforcement Training



Inverted Pendulum



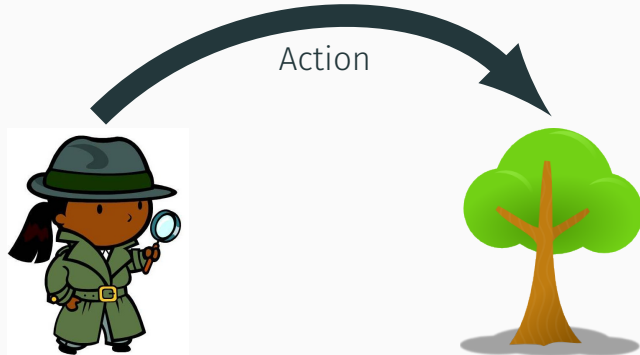
Reinforcement Learning

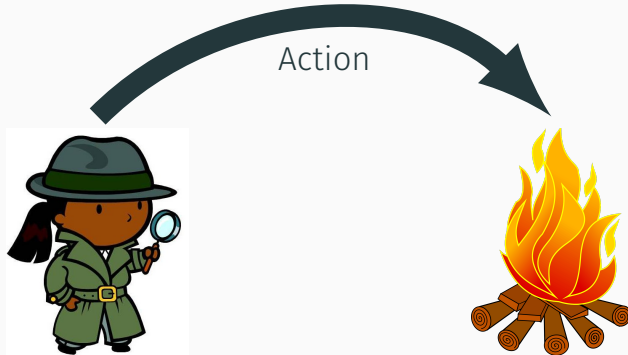


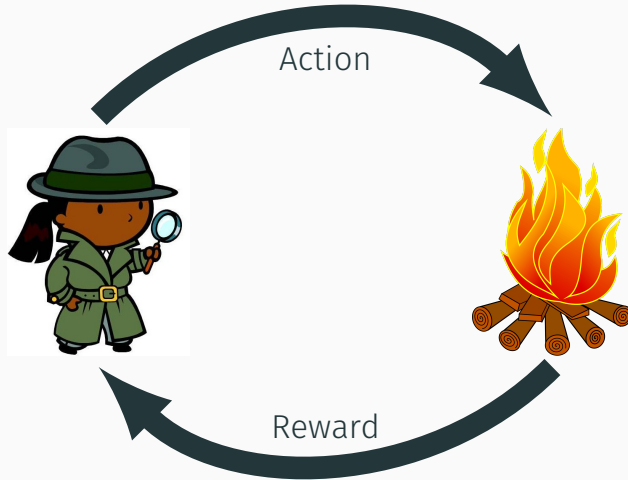
Can we learn without specifying how the task should be achieved by providing, rewards (positive) and punishment (negative)?



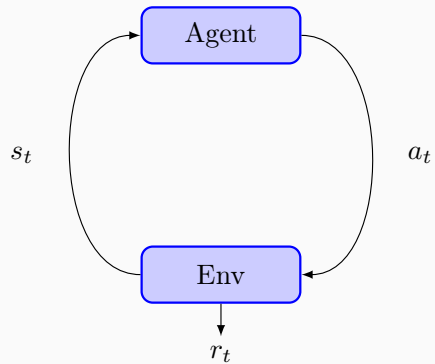


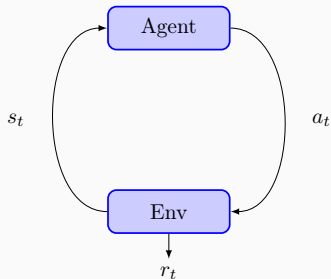






The Agent and the Environment





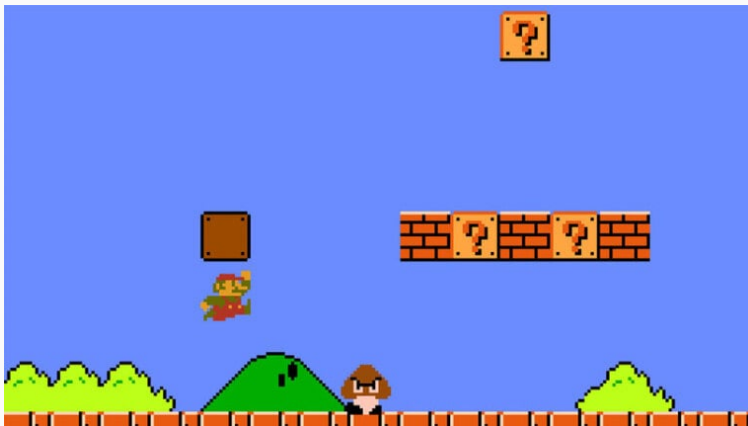
$$\operatorname{argmax}_{a_0, a_1, \dots} \sum_{t=0}^{\infty} r_t$$

- the agent takes actions in the environment
- actions change the state of the environment
- the state produces a reward

How is this different?

$$p\left(\sum_{t=0}^T r_t \mid s_0, a_0, a_1, \dots, a_t\right)$$

Lets just collect lots and lots of data and turn it into a regression problem



- When the number of possible sequences of input is very large

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- When acquiring data is hard and expensive

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- When its hard to define the "structure" of the problem explicitly as a cost

Markov Decision Processes

Definition (Markov Decision Process)

A Markov Decision Process is a 4-tuple (S, A, T, R)

S is the State space

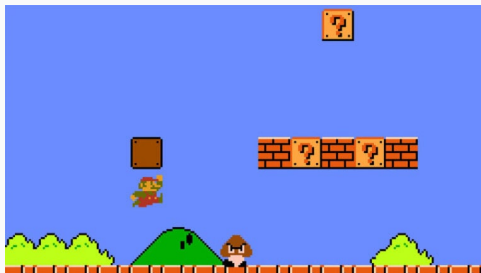
A is the Action space

T the state transition function

R the reward function

$$T(s_t, a_t) = p(s_{t+1} \mid s_t, a_t)$$

- RL provides solutions to Markov Decision Processes
- We must define our state space in such a way that the environment forms an MDP



State Space

- position and velocity of Mario
- the score
- number of coins collected
- time remaining
- Goomba position & velocity, state
-



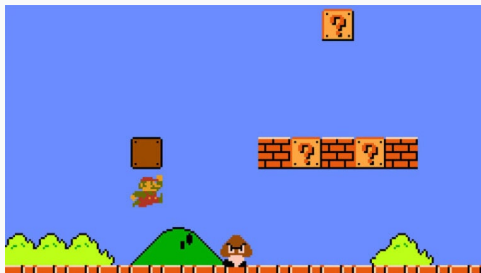
Action Space

- The controller has

$a, b, \uparrow, \downarrow, \leftarrow, \rightarrow$

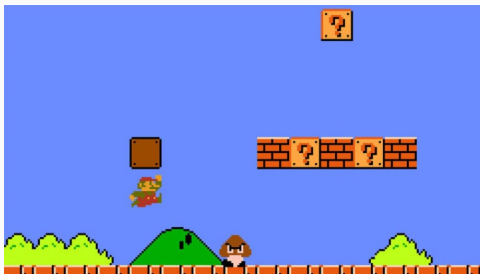
- Can press certain buttons together

$$A = \underbrace{\{0, 1, \dots, 8\}}_{\text{directions}} \times \underbrace{\{0, 1, 2, 3\}}_{\text{buttons}}$$



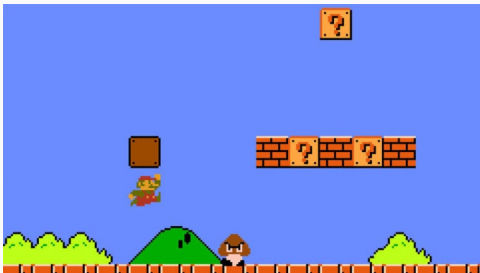
Transition Function

$$T : S \times A \rightarrow \Delta S$$



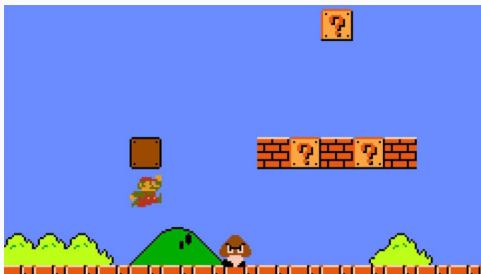
Reward Function

- beating the level (0, 1)



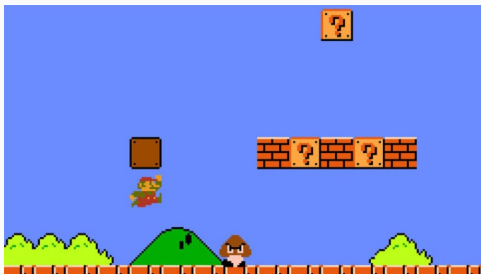
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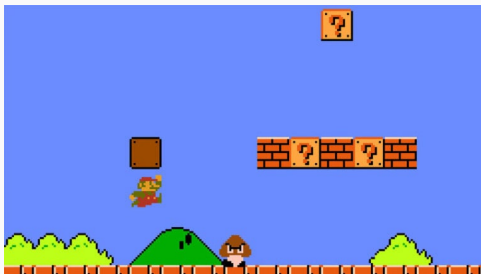
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- beating the level $(0, 1) + 0.01 \times \text{score}$



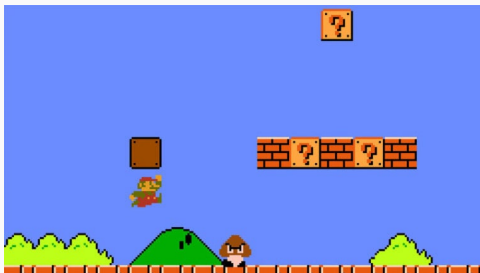
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 $(0, 1) + 0.01 \times \text{score}$
- beating the level/time

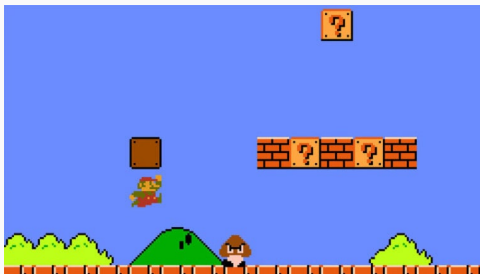


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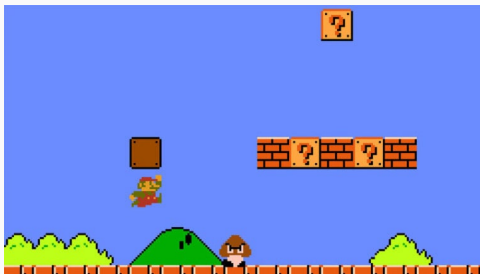
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- think carefully about constructing the environment



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- think carefully about how to structure reward signal



- think carefully about constructing the environment
- think carefully about how to structure reward signal
- now we have a problem to approach with RL

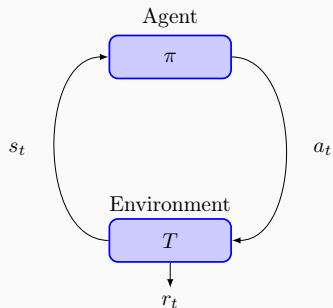
Reinforcement Learning

$$G = \sum_{t=0}^{\infty} r_t$$

- Task of RL is to design an agent to maximise return G

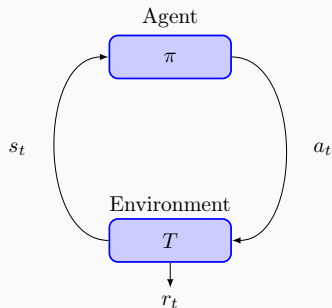
$$G = \sum_{t=0}^{\infty} \gamma^t r_t$$

- Task of RL is to design an agent to maximise return G
- We generally have a preference of cookies now compared to cookies later which we introduce by $\gamma \in (0, 1]$ called the discount factor



- The agent acts according to a policy π

$$\pi : S \rightarrow A$$



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$$\pi : S \rightarrow A$$

- Goal is to learn the optimal policy

$$\pi_* = \max_{\pi} \sum_{t=0}^{\infty} \gamma^t r_t$$

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$$p(s_{t+1} \mid s_t, a_t)$$

The Learning Problem

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- We have no data to learn from

What do we want to learn?



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- we want to interact with the environment as little as possible
- we only want to know the dynamics up to a level where the policy will act
- we only need to understand how to act to get a high reward
- learning the dynamic and policy are just proxies

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3. when we get data we can update our dynamics and policy
4. how to acquire data from the knowledge that we have?

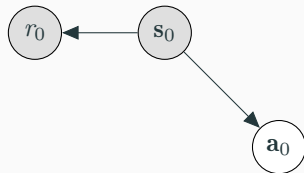
$$p(s_{t+1} \mid s_t, a_t)$$

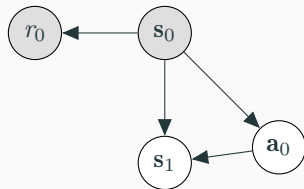
$$p(a_t \mid s_t)$$

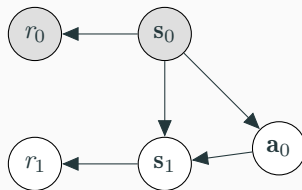
- Model the dynamics and the policy directly
- All the benefits of having uncertainty over each specific component
 - bias with prior knowledge
 - efficient exploration
- Just normal statistical inference :-)

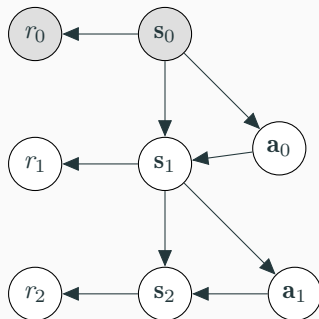




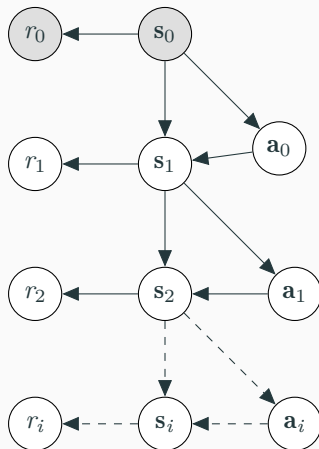




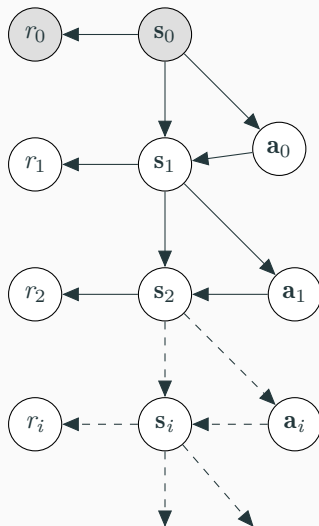




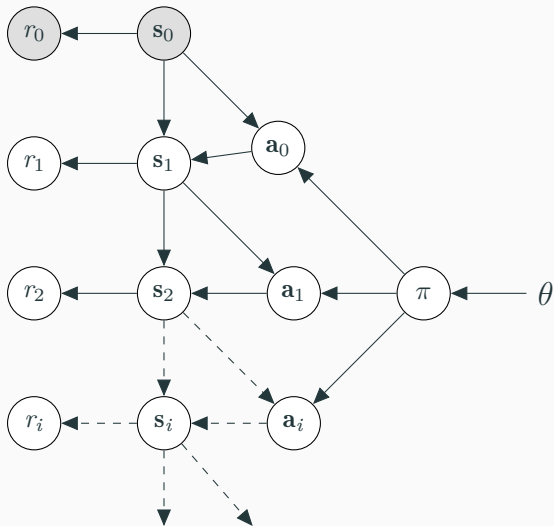
Reinforcement Learning



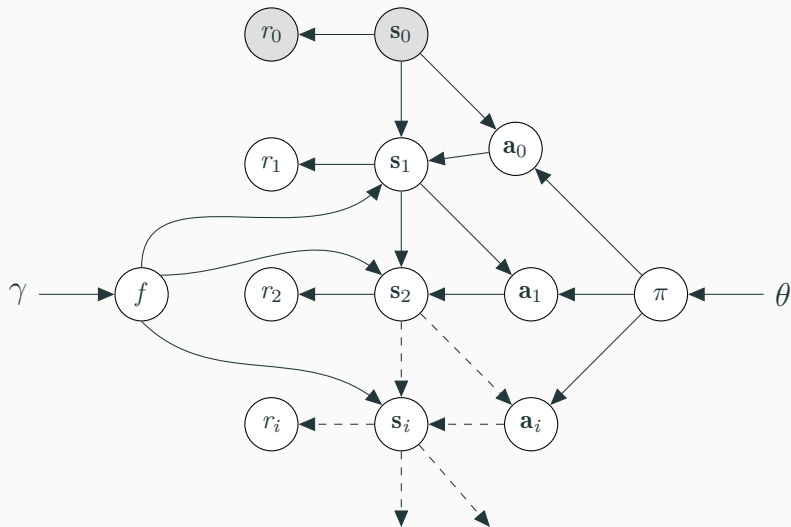
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Reinforcement Learning



$$\begin{aligned}
p(\mathbf{s}_{1,\dots,T}, a_{0,\dots,T-1}, r_{0,\dots,T}, f, \pi, \gamma, \theta | \mathbf{s}_0) = & \\
p(\mathbf{s}_T | a_{T-1}, s_{T-1}, f) p(a_{T-1} | s_{T-1}, \pi) \dots & \\
\vdots & \\
p(\mathbf{s}_2 | a_1, s_1, f) p(a_1 | s_1, \pi) & \\
p(\mathbf{s}_1 | a_0, s_0, f) p(a_0 | s_0, \pi) & \\
p(f | \gamma) p(\pi | \theta) p(\gamma) p(\theta) &
\end{aligned}$$

- if we want to learn dynamics and policy we need to marginalise out f, π, θ and γ
- very very hard problem as uncertainty propagated a long way

$$\mathbb{E}[r_t] = \int \int p(r_t \mid s_{t+1})p(s_{t+1} \mid s_t, a_t)p(a_t \mid s_t)da_tds_{t+1}$$
$$\pi_* = \max_{\pi} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid a_t \sim p(a_t \mid s_t) \right]$$

- We want to avoid explicitly learning a policy and a reward model
- Directly maximise the expected reward

- Define value function V ,

$$V_{\pi}(s_0) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid a_t \sim \pi(s_t) \right]$$

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- Lets move out the first action and define the function,

$$\tilde{V}_{\pi}(s_0, a_0) = \mathbb{E}[r_0 \mid a_0] + \mathbb{E} \left[\sum_{t=1}^{\infty} \gamma^t r_t \mid a_t \sim \pi(s_t) \right]$$

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- This we will refer to as the Q -function,

$$Q_{\pi}(s_0, a_0) = \tilde{V}_{\pi}(s_0, a_0)$$

- How should we pick our policy?

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$$\pi_*(s) = \operatorname{argmax}_{a \in A} Q_*(s, a)$$

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Lets re-arrange things

$$\begin{aligned} Q_*(s, a) &= \mathbb{E}[r_0 \mid a_0] + \mathbb{E} \left[\sum_{t=1}^{\infty} \gamma^t r_t \mid a_t \sim \pi_*(s_t) \right] \\ &= \mathbb{E}[r_0 \mid a_0] + \gamma \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_{t+1} \mid a_{t+1} \sim \pi_*(s_{t+1}) \right] \\ &= \mathbb{E}[r_0 \mid a_0] + \gamma \cdot V_*(s_1) \end{aligned}$$

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$$\begin{aligned}\theta_* &= \operatorname{argmin}_{\theta} \left(Q_{\theta}(s, a) - \left(\mathbb{E}[r_0 \mid a_0] + \gamma \operatorname{argmax}_{a_1 \in A} Q_{\theta}(s_1, a) \right) \right)^2 \\ &= \operatorname{argmin}_{\theta} \left(\underbrace{\left(Q_{\theta}(s, a) - \gamma \operatorname{argmax}_{a_1 \in A} Q_{\theta}(s_1, a) \right)}_{\Delta Q} - \mathbb{E}[r_0 \mid a_0] \right)^2\end{aligned}$$

- Can we learn a θ such that we can approximate the optimal Q ?
- For any $\{s_0, s_1, a_0, r\}$ we will receive an error

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 - with probability $1 - \epsilon$ take the action that maximises the Q
 - with probability ϵ take a random action

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 - allows for (more) principled acquisition strategies
 - very challenging inference task

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 - allows us to input our knowledge into the system
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 - very challenging inference task
- Model free RL
 - simple to implement and relatively cheap
 - hard to device principle acquisition strategies
 - often very data inefficient

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- We apply RL to problems where it is infeasible to acquire all data at the same time
- Just as BO the data-acquisition therefore has to be part of the inference mechanism
- There are many things where RL is a suitable approach

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 - or am I uncertain of the reward because I know what I should do (policy) but I don't know what will happen if I do (dynamic)?

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References

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